

DNA Replication & Repair - Factoid Sheet

Week 4 - Monday - Core Lecture 3 - current faculty deck + official Week 4 objectives

Core chain: replication fidelity → damage source → lesion type → repair pathway → disease when repair fails.

Replication Essentials

Semiconservative - S phase - bidirectional - eukaryotes use multiple origins.		DNA polymerase extends from a 3'-OH ; synthesis is 5' to 3' ; proofreading is 3' to 5' .
Factor	Job	
Pol III (bacteria)	Main replicase	
Pol I (bacteria)	Primer removal + repair	
Pol alpha	Primase associated	
Pol beta	Repair	
Pol gamma	Mitochondrial replication	
Pol delta	Lagging strand	
Pol epsilon	Leading strand	
PCNA	Euk sliding clamp/processivity marker	

Replication Mechanics

Helicase ATP-dependent strand separation.	Ligase ATP-dependent nick sealing.	RNase H Euk RNA-primer removal.
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Drug Targets

Mechanism	Examples	Outcome
No usable 3'-OH	Nucleoside analogs	Chain termination
Gyrase / topo II trapping	Ciprofloxacin, novobiocin	Bacterial DSBs
Topo II trapping	Doxorubicin, etoposide	DSBs
Topo I trapping	Camptothecin	SSBs

Topoisomerases

Type I One strand - one supercoil - ATP independent.	Type II Two strands - two supercoils - ATP dependent.
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Telomeres

End-replication problem: terminal RNA primer removal leaves no upstream 3'-OH. **Telomerase = RNA-dependent DNA polymerase**; RNA supplies template. Low in most somatic cells; active in stem cells and many cancers.

Fidelity / Processivity

3' to 5' exonuclease Proofreading removes a misincorporated terminal base.	Sliding clamp Holds polymerase at primer-template junction; euk clamp = PCNA.
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Damage Sources

Cause	Lesion
Replication errors / repeats	Mismatch, indel, repeat expansion
5-methyl-C deamination	C-to-T hotspot
Ionizing radiation	ROS, SSB/DSB
UV	Pyrimidine dimers
Intercalator	Insertion/deletion
Alkylator/crosslinker	Covalent base lesion / crosslink

DNA Repair - Fast Recall

General Repair Sequence

Recognize lesion → nick/remove damaged segment when required → DNA polymerase fills gap → DNA ligase seals backbone. Direct reversal is the important exception because the lesion is chemically reversed rather than excised.

Repair Pathway Matrix

Pathway	Fixes	Key step / association
Direct reversal	O6-alkylguanine	MGMT accepts alkyl group onto itself
MMR	Replication mismatch + small indel	MLH1/MSH2/MSH6/PMS2; Lynch; microsatellite instability
BER	Small altered base, deamination, oxidation, SSB	Glycosylase → AP endonuclease/lyase → pol/ligase
NER	Bulky helix distortion, UV dimer	Excise oligo around lesion; XP/Cockayne
HR	DSB with homologous template	Late S/G2; high fidelity; BRCA1/2
NHEJ	DSB without template	G1; end trimming/ligation; error prone
TLS	Replication-blocking lesion	Damage-bypass synthesis

MGMT Detail

MGMT removes O6-alkylguanine damage by transferring the alkyl group to a cysteine on itself. The enzyme is permanently inactivated ("suicide" transfer). High tumor MGMT can confer resistance to alkylating chemotherapy.

Disease Anchors

Lynch MMR defect → microsatellite instability → colorectal/endometrial + other cancers.	Xeroderma pigmentosum Global NER defect → severe UV sensitivity + skin-cancer risk.
Cockayne syndrome Transcription-coupled NER defect.	BRCA1/2 dysfunction Reduced HR of double-strand breaks.

Mutation Location

Location	Typical consequence emphasized
Intergenic / many introns	Often lower chance of direct protein effect
Regulatory region	Altered gene expression
Coding region	Altered amino-acid sequence/function or premature stop
Large DNA break/rearrangement	Can disrupt large amounts of information

Ames Test

His- Salmonella + test chemical on histidine-free medium. More revertant colonies = stronger mutagenic signal.

Tumor-Suppressor Logic

Repair-gene loss → mutation rate rises → genome instability → oncogenic + tumor-suppressor mutations accumulate. DNA-repair genes suppress cancer by preserving genome integrity.

Must-Not-Miss Traps

BER = bad base.	NER = bulky distortion.
MMR = replication mismatch.	HR/NHEJ = DSB.

30-Second Review

5' to 3' synthesis; 3' to 5' proofread. Topo I = SSB; topo II = DSB. Telomerase copies RNA into DNA. UV → dimer → NER. Lynch → MMR. BRCA → HR. Repair loss creates a mutator phenotype and cancer risk.

Assigned ScholarRx

DNA Replication: <https://exchange.scholarrx.com/brick/dna-replication> | DNA Mutation & Repair: <https://exchange.scholarrx.com/brick/dna-mutation-and-repair> | Trinucleotide Repeat Disorders: <https://exchange.scholarrx.com/brick/trinucleotide-repeat-disorders>